Eashan Patel

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EDUCATION

Rutgers University Honors College - B.S. in Computer Science, B.A. in Data Science **September 2022 - May 2026**

- Relevant Coursework: Data Structures, Computer Architecture, Design of Algorithms, Intro to Artificial Intelligence, Brain-Inspired Computing, Machine Learning, Regression Methods, Applied Statistics, Statistical Inference for Data Science

WORK EXPERIENCE

Software Engineer Intern **June 2024 – August 2024**

Voya Financial Remote

- Authored technical documentation for an **extJS codebase** maintaining client financial portals
- Developed flowcharts and Excel tracking systems to support the team's migration to a **new Salesforce cloud**
- Implemented **Java code** and created **4 new metrics** to track user interactions within retirement financial portals
- Collaborated with developers, analysts and stakeholders in an **Agile Scrum** environment

Online Instructor **June 2023 – August 2023**

iDTech Remote

- Taught **Python, Java, JavaScript and React Native** to **50+ students** in both group and one-on-one virtual classroom settings
- Collaborated with instructors and management to improve curriculum structure and student engagement
- Designed personalized lesson plans and adaptive teaching strategies to meet individual student needs

PROJECTS

ScarletParking | *JavaScript, Expo, React Native*

- Shipped a mobile application allowing Rutgers students to select parking passes and view valid parking locations via. interactive map functionality and real-time filtering
- Reached **Top 100** in Apple's Navigation category with **12,000+** impressions, **1,500+** downloads, and a **5-star** rating

RU Subletting | *TypeScript, Next.js, React, Tailwind, Supabase, PostgreSQL, REST APIs*

- Developing a full-stack web application where college students browse and post sublease listings through a marketplace
- Building the frontend with **Next.js, React and TypeScript**, including filtering by price, dates and amenities, detailed listing pages, and saved listings
- Designing a **PostgreSQL** schema on Supabase with **row-level security policies** for access control, integrating its **REST API** and writing custom server-side routes for server-only operations

Tandem | ***Meta AI Developer Hackathon*** **|** *TypeScript, Next.js, React, Python, FastAPI, PostgreSQL, MCP, Auth0*

- Developed a coordination platform where coding agents on separate machines work together without colliding, using **Meta's Muse Spark 1.3** to turn goals into chronological tasks, assign work by agent capability, and relay messages between agents
- Includes a **React**-based dashboard, a **FastAPI** backend, a **16-tool MCP server**, and a **PostgreSQL** schema, with user-login and machine connections secured using **Auth0 JWTs** and **OAuth 2.0 with PKCE**
- Built an optional **sandboxed runner** that uses **Anthropic Sandbox runtime, bubblewrap, socat and seccomp** to allow agents to continuously claim tasks, create Git branches, verify and push code, and open PRs

Robust Keyword Detection Using Spiking Neural Networks | *Python, PyTorch, snnTorch and Speech2Spikes*

- Created a **Spiking Neural Network (SNN)** using multi-layer **Leaky Integrate-and-Fire (LIF)** neurons with surrogate gradient training, which achieved **96% accuracy** on clean speech for binary keyword classification
- Designed experiments using noise distortions and afterwards demonstrated improved generalization, reaching **93.7% accuracy** on distorted audio while maintaining **>94%** clean-speech performance

Machine Learning Madness | *Python, XGBoost, scikit-learn, pandas, NetworkX*

- Built an **end-to-end pipeline** using Kaggle NCAA data to predict tournament point differential and win probability for men's and women's brackets, engineering 200+ matchup features from box scores, seeds and other rankings
- Adapted **Google's PageRank** into a margin-of-defeat-weighted strength metric that ranked among the top **XGBoost** features
- Found that combining team rankings and PageRank-based strength rating predicted tournament games more accurately than seed alone, cutting average prediction error to **8.9 points** for men's games (from 9.2) and **10.3** for women's (from 11.3)

SKILLS

- **Languages:** Python, JavaScript, Java, TypeScript, C, SQL, HTML, CSS, ExtJS
- **Frameworks & Tools:** React, React Native, Next.js, Node.js, FastAPI, PostgreSQL, Expo, Tailwind, Bootstrap, Vite, TanStack Query, Auth0, MCP, PyTorch, snnTorch, XGBoost, scikit-learn, NumPy, REST APIs, NetworkX, Matplotlib, Playwright, Render, Git/GitHub, Cursor, Claude Code
- **Core Development Skills:** Relational Databases, Data Structures & Algorithms, Machine Learning, Neural Networks, REST API Design & Integration, OAuth 2.0 / JWT Authentication, Full Stack Development, Mobile App Development, LLM Integration, Multi-Agent Systems, AI-Assisted Development, Test Automation, Technical Documentation, Agile/Scrum